

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (*BL2, BL5*)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

1. Rural Branch Office – Internet Connection, Devices and Security.

Introduction:

The rural branch needs Internet access for cloud applications, video conferencing, and secure communication with headquarters. Since wired high-speed Internet is unavailable and the budget is limited, the solution should be reliable, secure, and scalable.

Constraints:

- No high-speed wired Internet.
- Limited budget.
- Stable connectivity for video conferencing.
- Secure headquarters communication.
- Uninterrupted business operation.
- Future expansion.

Recommended Connection:

Use 4G/5G wireless broadband as the primary connection. If coverage is poor, satellite Internet can be used. A dual-WAN router can provide a backup connection and automatically switch when the primary link fails.

Networking Devices:

- Dual-WAN router – Internet connection, failover, VPN and traffic management.
- Managed switch – connects computers, printers, servers and access points.
- Wireless access point – provides office Wi-Fi.
- Firewall – blocks unauthorized traffic.
- UPS – keeps important network devices running during short power failures.

Security:

- Firewall protection.
- WPA2/WPA3 Wi-Fi security.
- Site-to-site VPN to headquarters.
- Multi-factor authentication.
- VLAN-based segmentation.
- Regular firmware and security updates.

Bandwidth and Cost:

QoS can give high priority to business applications and video conferencing. Wireless broadband reduces installation cost compared with dedicated wired infrastructure.

Scalability:

The network can later be expanded by adding switch ports, access points, bandwidth, VLANs, VPN connections, or a faster 5G/fiber connection when available.

Conclusion:

A 4G/5G dual-WAN design with firewall, managed switch, Wi-Fi, VPN, and UPS provides a practical balance of bandwidth, reliability, security, cost, and future scalability.

2. University Campus Wireless Network for More Than 3,000 Students.

Introduction:

The university needs campus-wide Wi-Fi for more than 3,000 students in classrooms, laboratories, libraries, and hostels. Because the number of access points is limited, placement and capacity must be carefully planned.

Constraints:

- More than 3,000 users.
- Many simultaneous connections.
- Limited access points.
- Low interference.
- Good coverage and performance.
- Secure authentication.
- Budget limitation.

Proposed Design:

Use a centrally managed wireless network: Internet → Router/Firewall → Core Switch → Distribution Switches → Wireless Access Points → Users.

Access Point Placement:

Place APs according to user density, building size, walls, expected traffic, and interference. Give priority to high-density classrooms, laboratories, libraries, and hostel areas.

Frequency Selection:

Band	Use
2.4 GHz	Longer coverage and better wall penetration, but more interference.
5 GHz	Higher speed, more channels, and better for high-density areas.

Prefer 5 GHz for high-performance users and use 2.4 GHz where longer range is needed.

Reducing Interference:

- Plan channels carefully.
- Avoid excessive AP overlap.
- Use suitable transmit power.
- Use automatic channel selection.
- Prefer 5 GHz where possible.
- Monitor wireless performance.

Authentication and Security:

Use WPA2-Enterprise or WPA3-Enterprise with a RADIUS server. Students, faculty, administrators, guests, and IoT devices can be separated using VLANs.

Group	Network
Students	Student VLAN
Faculty	Faculty VLAN
Administration	Admin VLAN
Guests	Guest VLAN
IoT	IoT VLAN

Performance and Cost:

Use capacity-based AP placement, load balancing, band steering, QoS, and centralized monitoring. Install APs where demand is highest and expand gradually to control cost.

Conclusion:

A centrally managed dual-band Wi-Fi network with careful AP placement, 5 GHz preference, enterprise authentication, VLANs, and monitoring provides good coverage, performance, security, and cost control.

3. Secure Network Architecture for a Financial Institution.

Introduction:

A financial institution must protect sensitive customer information while allowing employees to use business applications and online banking services. The design must provide strong security without seriously affecting performance or increasing cost.

Constraints:

- Sensitive customer information.
- Unauthorized access must be prevented.
- Employees need business access.
- Online services must remain available.
- Strict security and compliance requirements.
- Limited operational cost.
- Good network performance.

Proposed Architecture:

Use a layered architecture: Internet → Edge Router → Perimeter Firewall → DMZ → Internal Firewall → Core Switch → VLANs.

Network Segmentation:

Separate important systems into VLANs so that a security problem in one network does not easily spread to another.

VLAN	Purpose
10	Employees
20	Servers
30	Database
40	Management
50	Guest
60	Security/Monitoring

DMZ and Firewall:

Place Internet-facing servers in a DMZ. Keep internal databases away from direct Internet access. Firewalls should filter traffic, block unauthorized connections, control applications, support VPNs, and create security logs.

Authentication:

- Strong passwords.
- Multi-factor authentication.
- Role-based access control.
- Centralized identity management.
- Users should receive only the permissions required for their jobs.

IDS/IPS:

An IDS monitors suspicious activity such as unusual traffic, scanning, and unauthorized access. An IPS can also block detected malicious traffic.

Remote Access:

Remote employees should use a secure VPN with authentication, MFA, and encryption. Direct Internet access to internal systems should be avoided.

Performance, Compliance and Budget:

Use efficient firewall rules, QoS, suitable bandwidth, and monitoring to maintain performance. Maintain firewall, authentication, and IDS/IPS logs and conduct regular security reviews. Prioritize firewall, segmentation, MFA, IDS/IPS, backups, and monitoring to control cost.

Conclusion

A secure financial network should use VLAN segmentation, firewalls, a DMZ, MFA, role-based access, VPN, and IDS/IPS. This provides layered protection while balancing security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand